DO NOW — RECALL

Lesson 071 · Pipelining and Computational Methods Rev

Name: _____ Class: _____ Date: _____

1. What is a heuristic?

2. When does backtracking undo a choice?

3. What does throughput measure?

4. Rising-edge D flip-flop Q=1, D=0 immediately before active edge. What is Q after the edge?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Estimate guiding choices, not necessarily guaranteeing optimum.

2. Model answer

When the branch cannot produce a valid result.

3. Model answer

Completed items per unit time.

4. Model answer

Q becomes 0; D is sampled at the active clock edge.

By the end, you can…

• Pipelining overlaps separate stages of different tasks; it increases throughput after filling the pipeline, not necessarily single-task latency.

• Four tasks on a three-stage pipeline complete in six ideal cycles, versus twelve when each task runs all stages in isolation.

• Choose the computational method that fits: decomposition for parts; backtracking for reversible choices; heuristic for guided estimates; visualisation for patterns.

Vocabulary — know these terms

pipeline — Overlapping multiple processing stages.

throughput — Work completed per time.

latency — Time for one task to finish.

stall — Delay from dependency or resource issue.

NEW LEARNING

Read, annotate and explain

Observe

Pipelining overlaps separate stages of different tasks; it increases throughput after filling the pipeline, not necessarily single-task latency.

Explain

Four tasks on a three-stage pipeline complete in six ideal cycles, versus twelve when each task runs all stages in isolation.

Evaluate

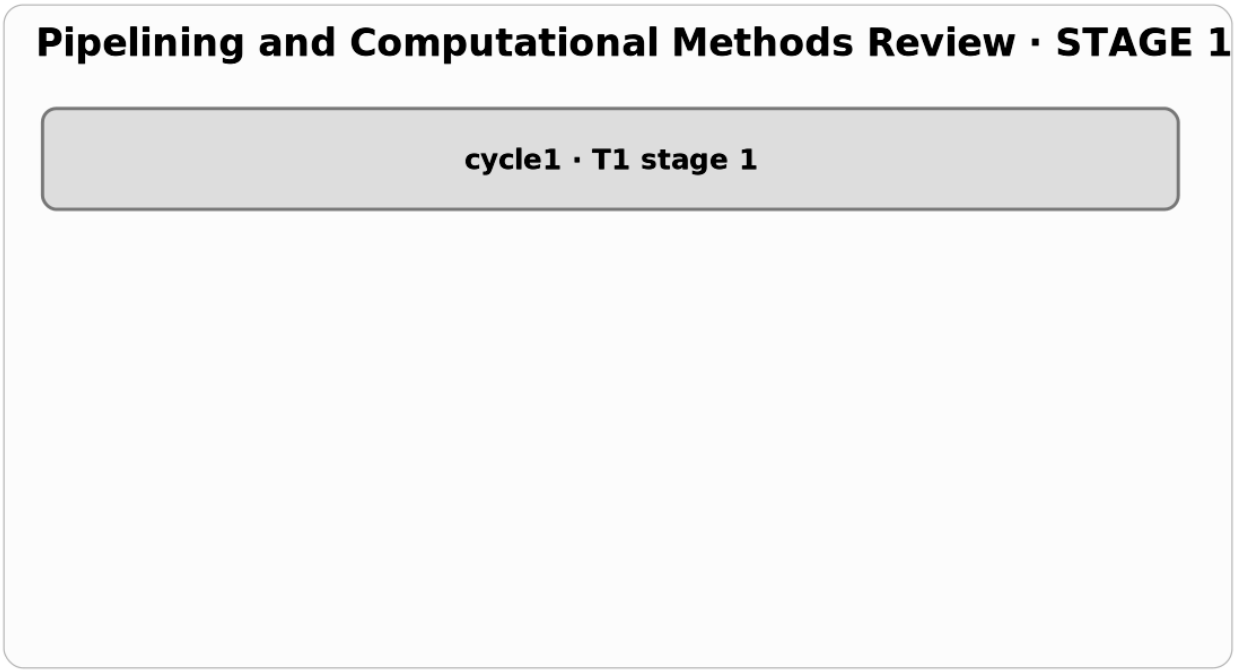
Choose the computational method that fits: decomposition for parts; backtracking for reversible choices; heuristic for guided estimates; visualisation for patterns.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

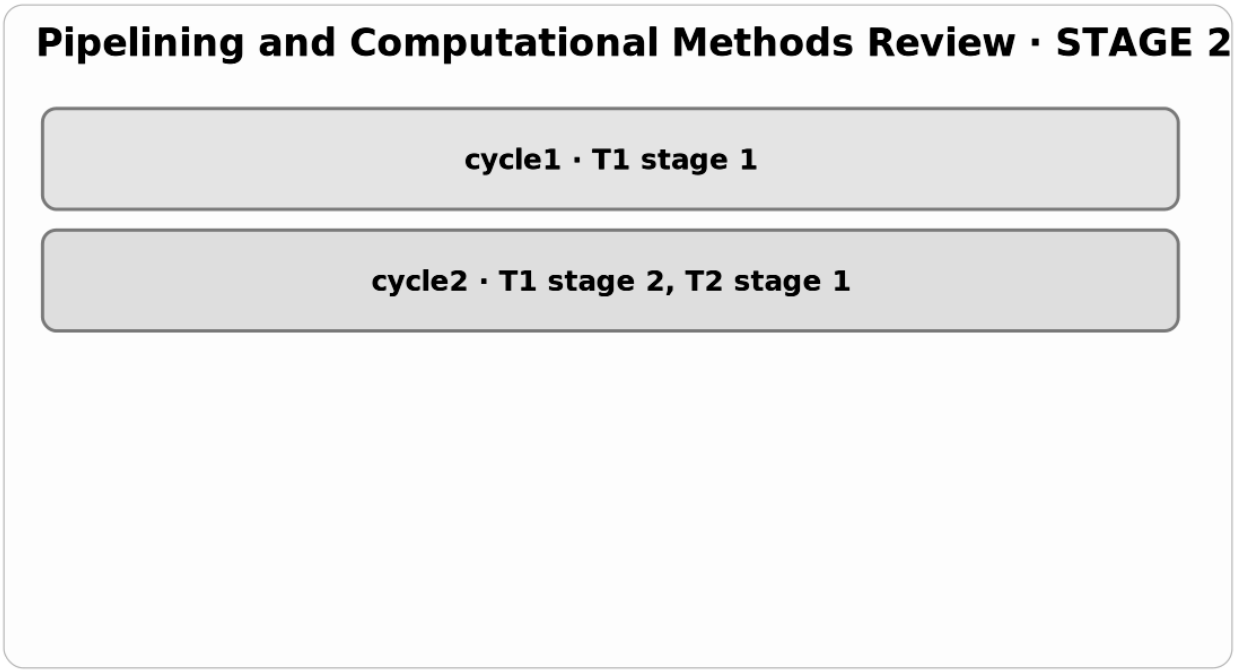


Pipelining and Computational Methods Review · STAGE 1: cycle1 · T1 stage 1

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

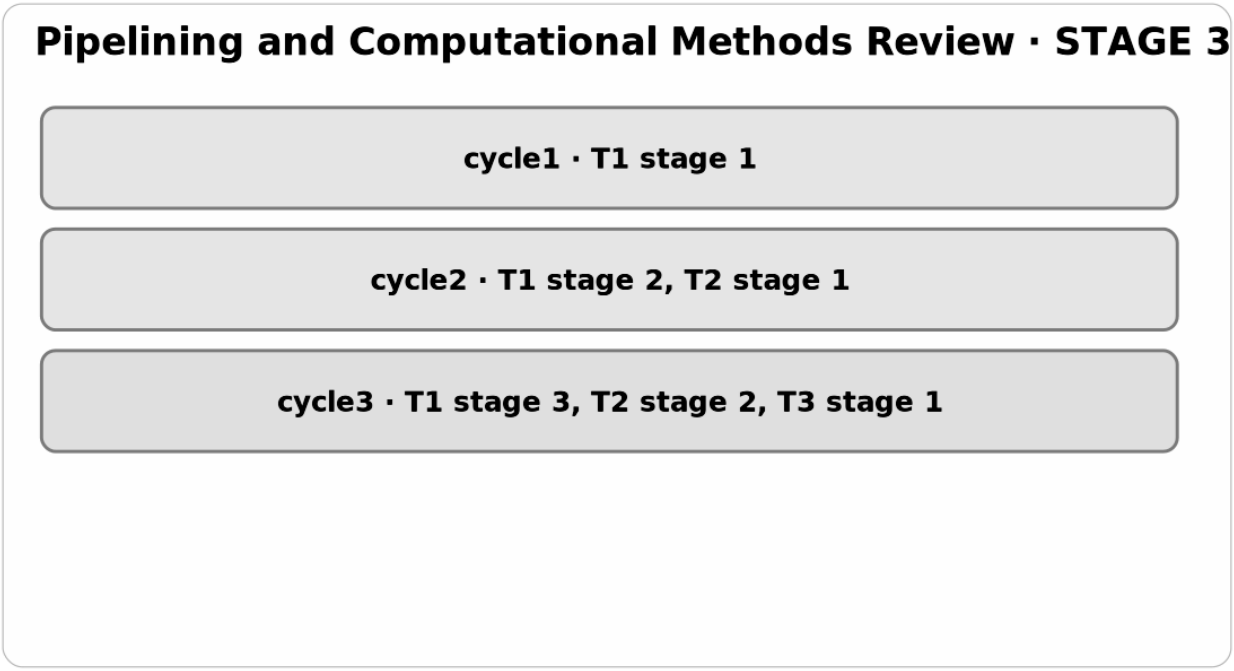


Pipelining and Computational Methods Review · STAGE 2: cycle1 · T1 stage 1; cycle2 · T1 stage 2, T2 stage 1

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

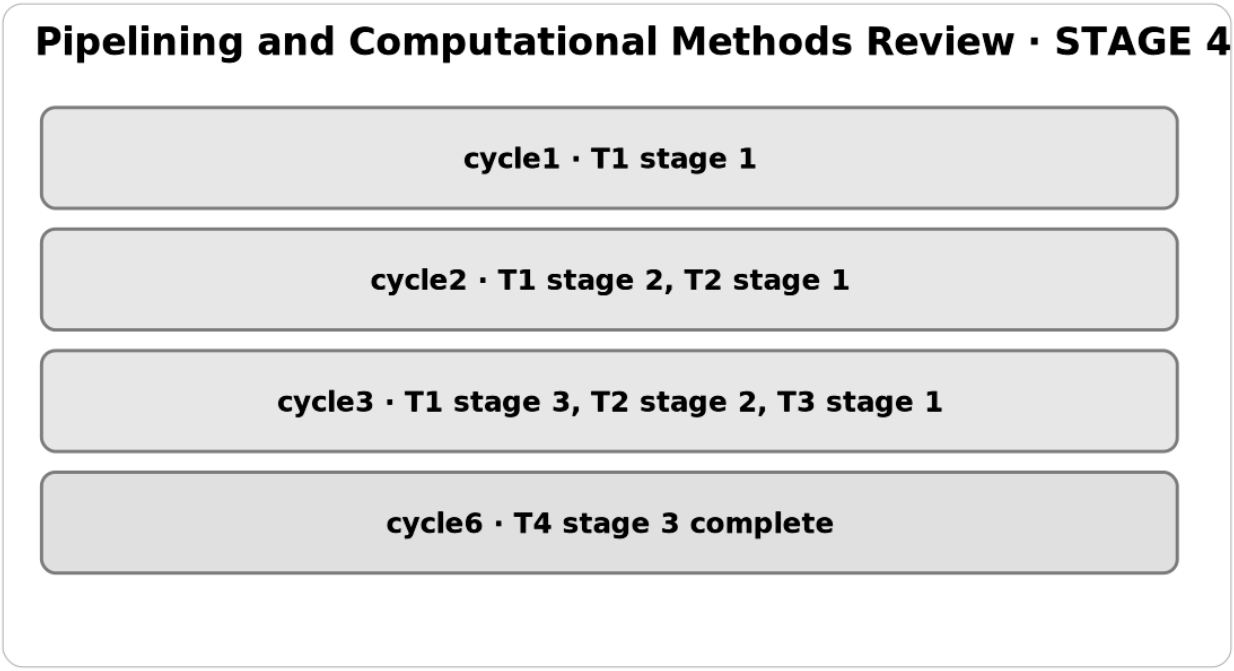


Pipelining and Computational Methods Review · STAGE 3: cycle1 · T1 stage 1; cycle2 · T1 stage 2, T2 stage 1; cycle3 · T1 stage 3, T2 stage 2, T3 stage 1

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Pipelining and Computational Methods Review · STAGE 4: cycle1 · T1 stage 1; cycle2 · T1 stage 2, T2 stage 1; cycle3 · T1 stage 3, T2 stage 2, T3 stage 1; cycle6 · T4 stage 3 complete

I DO — WORKED EXAMPLE

Follow the worked method

Find ideal cycles for 3 tasks through 2 single-cycle stages.

Explain

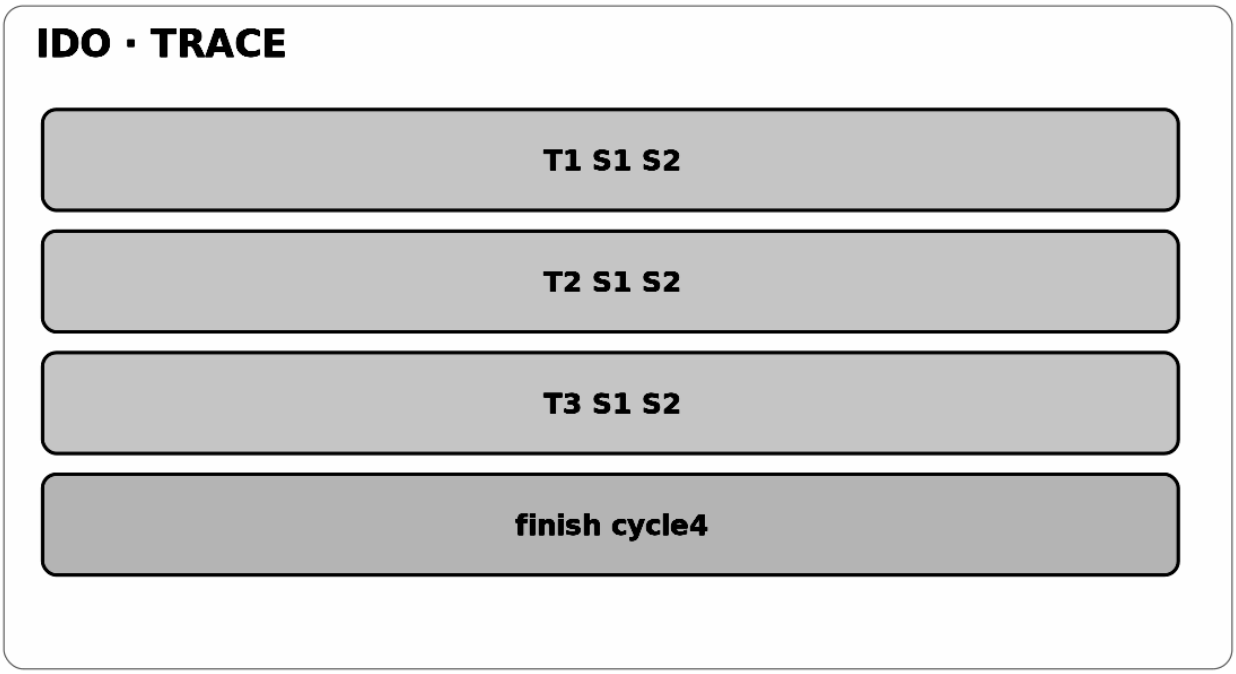
Start a new task each cycle while earlier tasks advance; last task finishes cycle4.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find ideal cycles for 3 tasks through 2 single-cycle stages.

Start a new task each cycle while earlier tasks advance; last task finishes cycle4.



IDO · TRACE: T1 S1 S2; T2 S1 S2; T3 S1 S2; finish cycle4

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

For 5 tasks with 3 equal-length pipeline stages, predict ideal cycles.
Source code / input:
3+5−1=7 cycles; sequential total=15 cycles.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Draw ideal 2-stage pipeline schedule for 3 tasks and state completion cycle. [6]

Question 2

For 6 tasks and 4 one-cycle stages, compare pipelined and sequential ideal times. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Match backtracking, data mining, visualisation and pipelining to four problems: maze with dead ends, shopping patterns, charting results and overlapping video processing. Explain each. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A pipeline always divides each individual task's latency by its number of stages.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

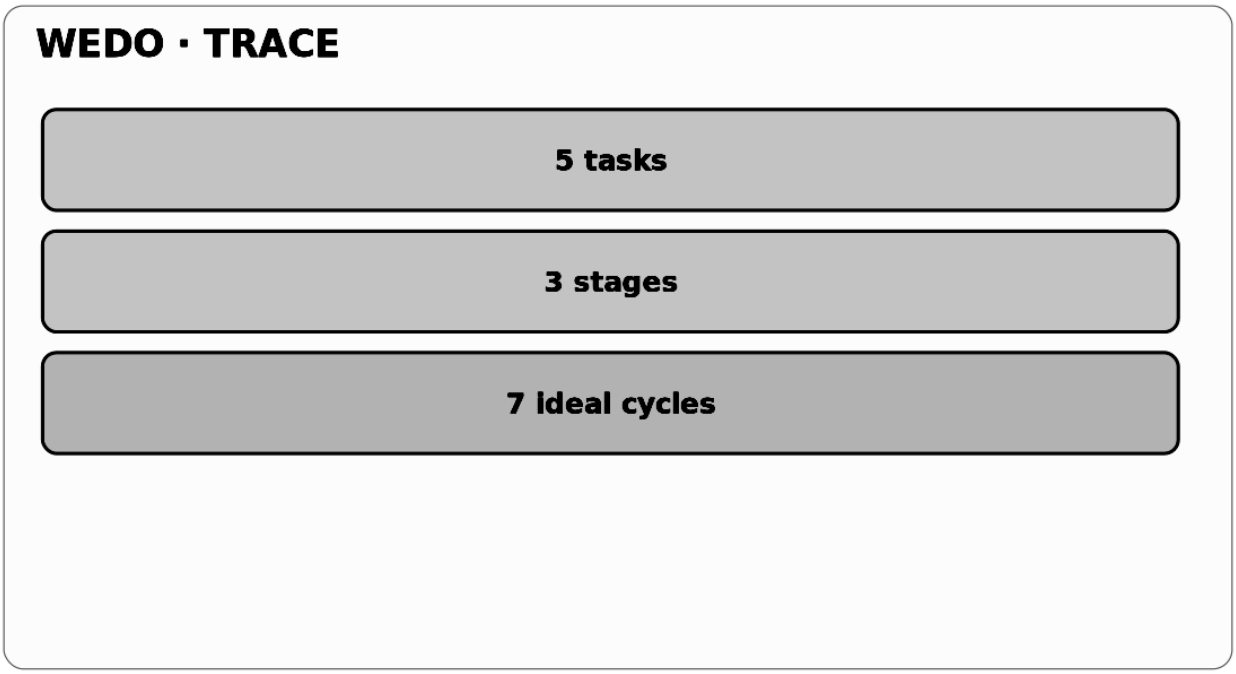
Explain
This is a throughput model assuming one task enters each cycle and no stall.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For 5 tasks with 3 equal-length pipeline stages, predict ideal cycles.

This is a throughput model assuming one task enters each cycle and no stall.



WEDO · TRACE: 5 tasks; 3 stages; 7 ideal cycles